The reversibility test — information as output, not as a base unit

The bit as the ultimate reduction (Doc. 301)

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Doc. 301 The reversibility test — information as output, not as a base unit

.1 Occasion and scope

The collaboration produced a stability picture that carries three quantities — information, coherence and energy (ICE) — as equally ranked corners of a triangle, with a reference at the centre. This document examines what categorical standing "information" actually occupies in such a picture. The finding is directed but not single-level: *information* is not synonymous with *bit*. The term splits into three levels — the bit (comparator output), the description (chosen representation) and the absolute identity (the complete object itself). As a base unit beside energy, information qualifies on neither of the two lower levels; it is fundamental only in the absolute sense, as the topmost identity. The criterion that carries this classification is reversibility. Scope: a Category-B finding (categorical classification, no physical derivation, no claim that FFGFT derives a bit). The result is a sharpening of the reduction principle of Doc. 296 and the comparator reading of Doc. 300.

.2 The criterion: fundamental means reversible

A quantity is fundamental exactly when its transformation has no distinguished direction — when one can turn it on its head and it still stands. The touchstone is energy in natural units. Through the relation

$$\tilde{T} \cdot m = 1 \tag{.1}$$

energy, mass and the dual time are not independent axes but three views of one quantity:

$$E = m = \frac{1}{\tilde{T}}. \tag{.2}$$

This transformation is symmetric: one may begin at E and read off m, at m and read $\tilde{T}$, at $\tilde{T}$ and read E. No side is the primary one, none is input or output. This is precisely why energy is intrinsic and the energy scale E_0 belongs to the framework natively (Doc. 292): the transformation discards nothing, each view carries the others fully. Here reversibility is not an addition but the expression of the fact that nothing is lost. In SI, $\hbar$ and c re-enter as conversion factors and pull the three apart; that is the projection, not the core.

.3 Intrinsic and relational

Not every quantity carries itself. A geometric quantity is **intrinsic**: an area, a winding has its value without reference to anything else. A thermodynamic or informational quantity is **relational**: entropy is defined only relative to a coarse-graining, a bit only relative to a distinguished reference. There is no bit "as such". This distinction is the categorical basis of what follows: intrinsic quantities can be fundamental, relational ones always presuppose a *with respect to* and therefore cannot be self-carrying.

.4 The ICE triangle as a comparator

Read in circuit terms, the ICE picture coincides corner for corner with the comparator of Doc. 300 — not by analogy, but as the same object drawn from the other side. Usable information at the output of a comparison requires three ingredients and a fourth condition:

- **Energy** — the signal at the $+$input carries a level; without a level there is no comparison.
- **Coherence** — the signal has phase structure, is not mere noise; geometrically the phase on the windings.
- **Reference** — the $-$input, the threshold against which the decision is made; it must come from a *different*, derived source than the signal (Doc. 300: reference from derivation, not from a copy).
- **Hysteresis** — a bare threshold chatters at the switching point; a transition counts only once it stands over a dead-band (amplitude hysteresis) or persists across

a window (time hysteresis, debouncing). Without a dead-band the output is meaningless, however clean the energy and coherence.

Important: The ICE triangle contains energy, coherence and reference, but no declared dead-band. Hysteresis is the fourth condition without which "threshold exceeded" does not become "transition counts". The time hysteresis used in the finite proxy implementation is exactly the sliding window M_m (Doc. 300/297) — a debouncer, and as such a short-term buffer, not an accumulating memory.

.5 Information is output, not input

The comparator thus inverts the order the triangle suggests. Energy, coherence, reference and hysteresis stand at the *input*. Information is what emerges at the *output* once those conditions are met: the bit the threshold decision produces. It is not the ingredient the comparison is built from, but its result. But a quantity that presupposes the whole apparatus in order to arise cannot be the base unit the apparatus is made of. This is also why earlier attempts to anchor a bit as a base unit had to fail: they sought to posit the output product as an input quantity — a circularity.

.6 The reversibility test

The criterion of Section 2, applied to information, comes out negative. E–m–$\tilde{T}$ is bidirectional: each view reconstructs the others, because nothing is discarded. The comparator is *unidirectional*:

$$\text{signal} + \text{reference} + \text{hysteresis} \;\longrightarrow\; \text{bit}, \tag{.3}$$

and this arrow does not reverse. From a bit one does not recover the signal, not its level, not its phase, not the reference threshold against which the decision was made. Fundamental quantities are reversible because they preserve; information is not reversible because it erases. *Fundamental and bit thus exclude one another not accidentally but by definition*: reversibility demands that the generating structure remain preserved in the result, and the bit is precisely the operation that does not preserve it.

.7 The bit as ultimate reduction

By Doc. 296, every evaluation is a reduction: it leaves out, and precisely by leaving out it makes a sub-domain evaluable. The bit is the limiting case of this statement — the *maximal* reduction. A whole signal course, with its energy, its phase, its position relative to the threshold, is collapsed to a single yes/no; less is not possible. And the irreversibility is not an addition to the reduction, it *is* the reduction: the threshold, the phase, the input conditions *were* — at the moment the comparator flips, they are gone. The bit no longer carries how it came about. It says "above" or "below", not at which level, not at which phase, not how far from the reference. Everything that produced the result is no longer contained in the result — and therefore there is no way back, because the where-to-return-to is exactly what the reduction has erased.

.8 The Landauer connection

This irreversibility is the thermodynamic side of the same matter. Erasing a bit costs $k_B T \ln 2$ and produces entropy; the direction is built in, because erasing information produces heat. To reverse the comparator would mean recovering that entropy, and that is thermodynamically excluded. This explains in retrospect why anchoring a base-unit bit to thermal motion went so badly in the prior working state: the attempt bound a directed, reducing quantity to a reversible base structure; the directions do not match. Not a lack of calculation, but a categorical incompatibility was the cause.

.9 Three levels of information: bit, description, absolute identity

The reversibility test separates cleanly, but the boundary does not run at “information” wholesale. The term splits into three levels that answer the test differently.

First level — the bit. The comparator output: reference-dependent, generated at the output, erasing the input conditions. It fails — unidirectional, reducing, the ultimate reduction (see “The reversibility test” and “The bit as ultimate reduction”). A consequence, not a foundation.

Second level — the description. Information is not the bit but, first, a *description*: the representation of a state of affairs. Which kind of description is chosen — geometric, mathematical, rhetorical — changes the representation, not the state of affairs. This is the same line as in Doc. 297/300: a representation is a choice of coordinates, not a structural property of the object; different descriptions of the same object are mutually translatable, but none is the object. And none is ever *complete*: every description is a projection (Doc. 296), hence a reduction, hence it leaves out. A complete description would exist only if a projection left nothing out — and then it would evaluate nothing. The description therefore also fails the reversibility test: as a route back to the object no single description is invertible, because each is incomplete. The descriptions are mutually translatable, but none reconstructs the whole.

Third level — the absolute identity. Fundamental in the full sense is the complete object itself alone: the boundaryless T^4 object (the core ξ on T^4), of which every projection shows only a section. *Information in the absolute sense* is not a quantity *in* the system and not a single description, but the complete self-identity of the object with itself, prior to any projection. Only this level passes the reversibility test — trivially, because there is nothing to leave out: there is no “with respect to”, because there is no external standpoint. This is the topmost identity; to it, and to it alone, “information” belongs fundamentally (self-reference, Doc. 248 §8).

Important: The geometric structure of the core — winding numbers as topological invariants, carried losslessly — belongs to the third level, to the object itself; it is not "information" in the bit sense. A geometric *description* of that structure, by contrast, remains second level: a projection. And the bit read off is first level. The three meanings must not be conflated. In particular: if anything stands "above" energy, it is not the bit but the complete object — and energy is one of its projections (a particularly intrinsic one, but a view).

.10 Digitisation: the staircase and its smoothing

The reason a bit can never carry the whole information is the basic mechanism of digital technology: analogue-to-digital and digital-to-analogue conversion. A bit is the 1-bit quantisation — the coarsest possible staircase, a single threshold. Any finer staircase with finitely many steps remains categorically the same object, only at higher resolution.

The torsion winding provides the natural example. The unrolled recursion closes after exactly 75 equal segments (Doc. 300/295); decomposing the winding into these 75 parts digitises it — 75 discrete steps. By DA conversion a continuous course can be regained from them. But the reconstruction is already lossy, for two reasons:

- Quantisation discards what lay *between* the steps. The continuous structure between the 75 nodes — the exponentially shrinking steps of the log-spiral, the cumulated integral $D(k)$ (Doc. 295) — is not contained in the 75 values.
- The DA converter outputs a staircase, not a smooth curve. One must *smooth* the steps (reconstruction filter). But the smoothing is an interpolation — an assumption the filter adds, not information contained in the digital value. What lies between the steps is guessed, not recovered.

The AD/DA round trip is thus precisely *not* a true inverse but a directed, lossy process: the reversibility test in circuit form. The lost intermediate structure is not reconstructed by the smoothing but replaced by a filter assumption — the same signature as the bit, whose input conditions "were" and no longer stand in the result.

Important: Even the *natural* staircase — 75, the closure number — does not carry the continuum. It is the right number for the closure step, but not for the continuous course: the full object is richer than any finite staircase (Doc. 296). No N, not even the structurally distinguished $N = 75$, carries the continuum. Placed within the three levels: the bit is the 1-step quantisation (first level, output); any finite N-staircase — including the 75-division — is a description of finite resolution (second level), finer than a bit but still lossy, still a projection; the continuous winding, the complete object, alone carries everything (third level). The smoothing is exactly the place where the description overlays its incompleteness with an assumption.

.11 Refinement: resonances enforce the quantisation

The previous section spoke of the "continuous structure between the nodes". This must be sharpened, for on the finest scale the object is not continuous. The torus is compact, and a compact torus has a *discrete* spectrum: the resonances on T^4 — the eigenmodes of the T^4 connection-Laplacian (Doc. 283) — enforce a quantisation. In the smallest parts *not all possibilities are equally weighted*; the resonance structure singles out particular discrete states. This is exactly why the Standard Model comes out discrete — the masses, $\theta = 2/9$, the Z_3 structure are resonances, not arbitrary points of a continuum.

From this follows a restriction of the loss argument: there is a *smallest possible jump*, the quantum set by the resonances. A conversion that goes down to this smallest jump and meets the resonance structure is *permitted* — it is not lossy, because below the jump there are no further states that could be lost. What lies between a *coarser* staircase is therefore not a continuum but finer *discrete* resonance structure, and it has a floor.

The reversibility test thus sharpens: loss does not arise from quantisation as such, but only from a staircase that is *coarser* than the resonances or lies *across* them — for instance an arbitrary equal division. A resonance-faithful conversion down to the smallest jump passes the reversibility test: it captures the actual discrete structure, not an approximation. The bit nonetheless remains the ultimate reduction — a single threshold is coarser than the resonance floor and does not meet it. And beyond a single scale it still holds that no description of one scale captures the cross-scale richness of the full object (Doc. 296). The statement "the full object is richer than any finite staircase" is therefore to be read *across scales* — within a scale, the resonance-faithful staircase is by contrast the true structure.

.12 Temporal and topological boundarylessness

Two properties of the absolute identity follow from analysis and topology and are therefore booked as *consequences*, not as posits.

No reachable edge of the recursion. The map $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ has $\xi = 0$ as a fixed point; ξ_n decays monotonically towards zero but reaches zero in no finite step (Doc. 296). There is no first and no last step. A "simplest initial state" — a scale without fractal structure, from which the refinement would only begin to grow — does not exist: the self-similarity (factor $1/e$ per 2π turn, the log-spiral) is not a property that builds up but is already fully present in the map. A structural beginning would break the scale invariance that defines the recursion — a contradiction, not a world-view.

Time without a markable origin. Experienced time *is* the cumulated defect $D(k) = \sum 100\,\xi_n$ (Doc. 295), the integral over $\delta t = 1/(100\,\xi_n)$. Since the recursion has no first step, $D(k)$ has no distinguished origin. Time is the recursion, not its container; to seek a "beginning of time" would place the recursion at a moment it only generates.

No external standpoint. T^4 is compact and boundaryless, $\partial T^4 = \varnothing$. That is a topological fact, not a posit. From it follows that there is no standpoint outside the object — and hence the "no with respect to" of the absolute identity: the complete object has no outside against which to demarcate itself.

Taken together: the absolute identity is boundaryless spatially (no outside, $\partial T^4 = \varnothing$) and temporally (no beginning, no reachable edge of the recursion) — the same property in two guises: no standpoint beyond the object. These statements are derivable from analysis and topology and carry no speculation marker.

.13 The reflexive case — what goes beyond the mathematics

What follows explicitly goes beyond the mathematics and is booked as philosophically speculative (P35). These are two interpretive jumps, not calculations.

Applying the three levels to FFGFT itself, the framework — the torus, $\tilde{T} \cdot m = 1$, ξ across all scales — is a *description* (second level), necessarily incomplete. This is not a weakness of this framework in particular but what a description is: no theory is ever its object, each projects and leaves out. FFGFT says so explicitly, whereas naive realisms behave as though they were the territory.

First jump — the realist posit. That the resonance-faithful description traces a *real* object and is not merely a consistent structure is posited, not proven. The mathematics shows resonance fidelity (α at the measured value, the masses as eigenmodes, Koide); that fidelity traces reality rather than mere consistency is the realist wager. The deflationary reader may always say: all just description, the object a useful fiction. FFGFT's answer is not a proof but the wager that the resonance-faithful success is better explained by a real object than by coincidence — and the only distinctive thing is that the resonance hits make that wager falsifiable.

Second jump — the diagnosis of the Word reflex. The pull of the information-as-fundamental idea ("In the beginning was the Word"; "it from bit"; the mass-energy-information equivalence) may stem from a deep cultural reflex: that information is the first substance from which matter follows. The reversibility test inverts the order: the word is the bit, and the bit is the last reduction, not the first creation. In the beginning was the geometry (the object); the word is what one of its scales reads off. This is a cultural-cognitive hypothesis about *why* the intuition pulls — not a mathematical statement; the reversibility test itself remains mathematical.

Self-projection. FFGFT is a T^4 process describing T^4 (Doc. 248 §8): the description is of the same stuff as the object and is generated by it (the describing process is itself a T^4 process on the cognitive scale). FFGFT is therefore neither external to the object nor the object, but its *self-projection* — the object describing itself on one of its own scales. Self-reference as an ontological locus, booked as a posit.

Hence the limit of the opening question: "How much ontological reality does FFGFT have?" is itself a second-level question — it presupposes a describing standpoint that does not exist over against the third level. One cannot ask from outside how much reality the object has, because there is no outside; one cannot ask what came before it, because there is no before. At the level of the absolute identity the question dissolves.

.14 Status

Categorical classification (Category B), not a physical derivation step. The finding places information relative to programmes that posit it as a physical base quantity (such as a mass-energy-information equivalence): such programmes treat information as input; the reversibility test shows that the bit is output. The three levels are to be booked at different temperatures:

- **Bit** — *proven*: the ultimate, input-condition-erasing reduction is irreversible and therefore a consequence, not a base unit (reversibility test).
- **Description** — *structural*: every representation is a projection and hence incomplete (reduction principle, Doc. 296); a choice of coordinates, not a structural property of the object.
- **Absolute identity** — *ontological posit*: information in the absolute sense is identified with the complete object and is the topmost identity (self-reference, Doc. 248 §8). This is posited, not derived.

The load-bearing statement in one sentence: *fundamental ⇔ reversible; bit and single description fail the test (the bit erases, every description is incomplete), the absolute identity alone passes it — so information counts as fundamental only in the absolute sense, as the topmost identity, and this is in FFGFT the core ξ on T^4, not a quantity beside it*. Connections: Doc. 296 (reduction as the condition of evaluability), Doc. 300 (comparator, reference, hysteresis), Doc. 283 (discrete resonance spectrum on T^4), Doc. 292 (E_0 intrinsic), Doc. 248 §8 (self-reference).